7. Generation of a sine or cosine wave using DDS compiler IP of Xilinx Vivado

(a) Generation of a sine wave with a frequency of 5MHz

Aim

To design and simulate a Direct Digital Synthesizer (DDS) using the DDS Compiler IP core for the generation of a 5 MHz sine wave signal.

Software Required

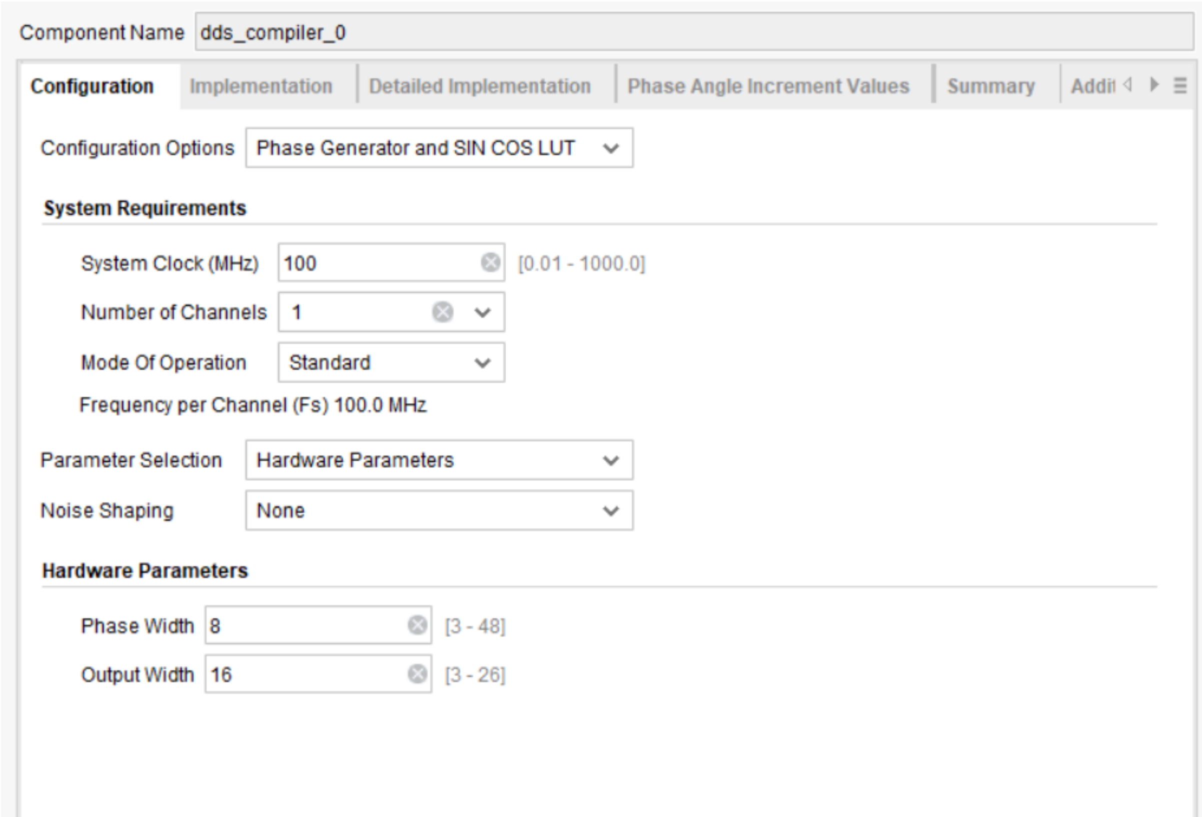
- Xilinx Vivado Design Suite
- Vivado Simulator
- Vivado IP catalog

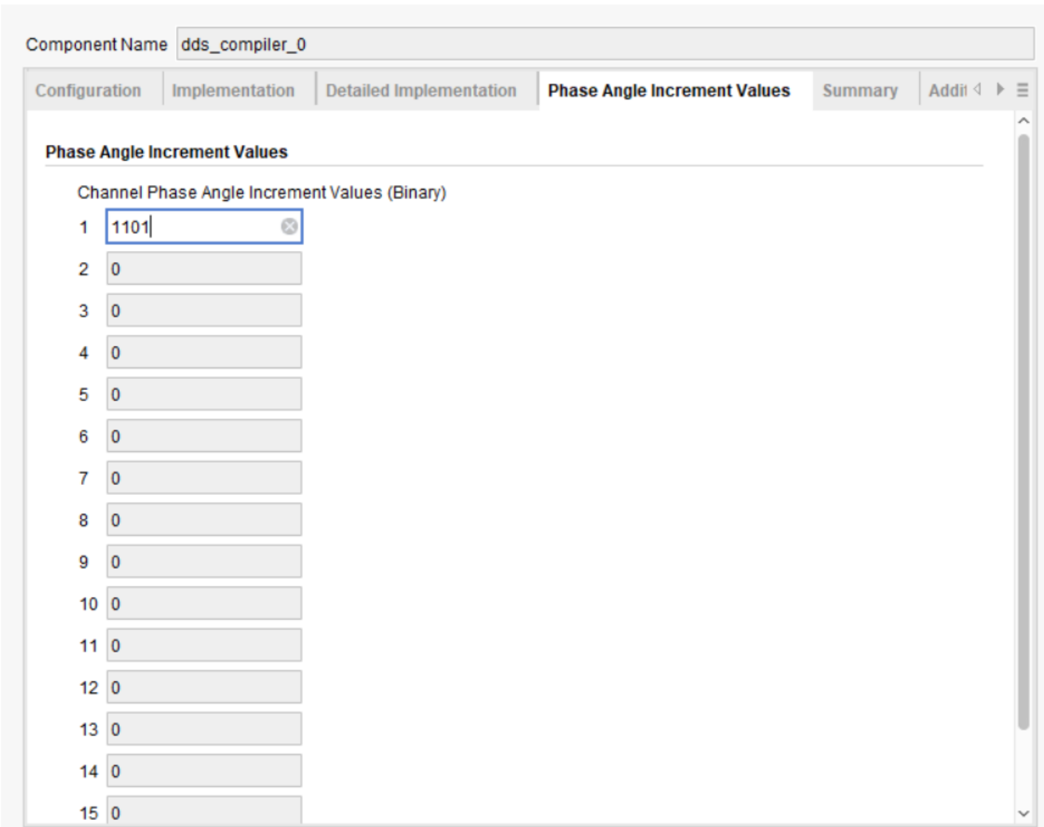
Procedure

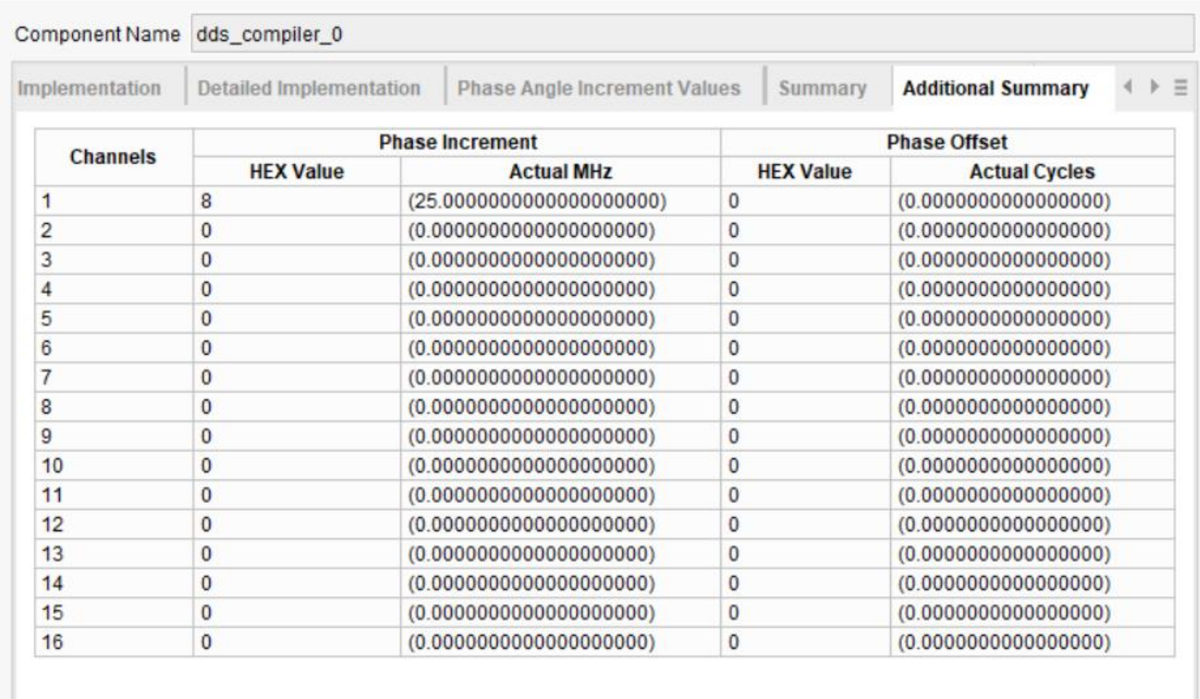
1. Open **Xilinx Vivado** and create a new RTL project.
2. Select the target FPGA device/board.
3. From the **IP Catalog**, add **DDS Compiler (dds_compiler_0)**.
4. Configure the IP as follows:
    - Configuration: *Phase Generator and SIN COS LUT*
    - System clock: **100 MHz**
    - Channels: **1**
    - Mode: **Standard**
    - Phase width: **8 bits**
    - Output width: **16 bits**
    - Noise shaping: **None**
5. Go to the **Phase Angle Increment Values** tab and set:
    - **Phase Increment = 13**
6. Generate the IP output products.
7. Write a testbench to apply clock and reset and to capture sine and cosine outputs.
8. Run behavioral simulation and observe the output waveforms.

fout=(Phase Increment×fclk) / 2^(Phase width)

$$f_{out} = \frac{13 \times 100\,\text{MHz}}{256} \approx 5.08\,\text{MHz}$$


Component Name
dds_compiler_0
Configuration
Implementation
Detailed Implementation
Phase Angle Increment Values
Summary
Addit
Configuration Options
Phase Generator and SIN COS LUT
System Requirements
System Clock (MHz)
100
[0.01 - 1000.0]
Number of Channels
1
Mode Of Operation
Standard
Frequency per Channel (Fs) 100.0 MHz
Parameter Selection
Hardware Parameters
Noise Shaping
None
Hardware Parameters
Phase Width
8
[3 - 48]
Output Width
16
[3 - 26]

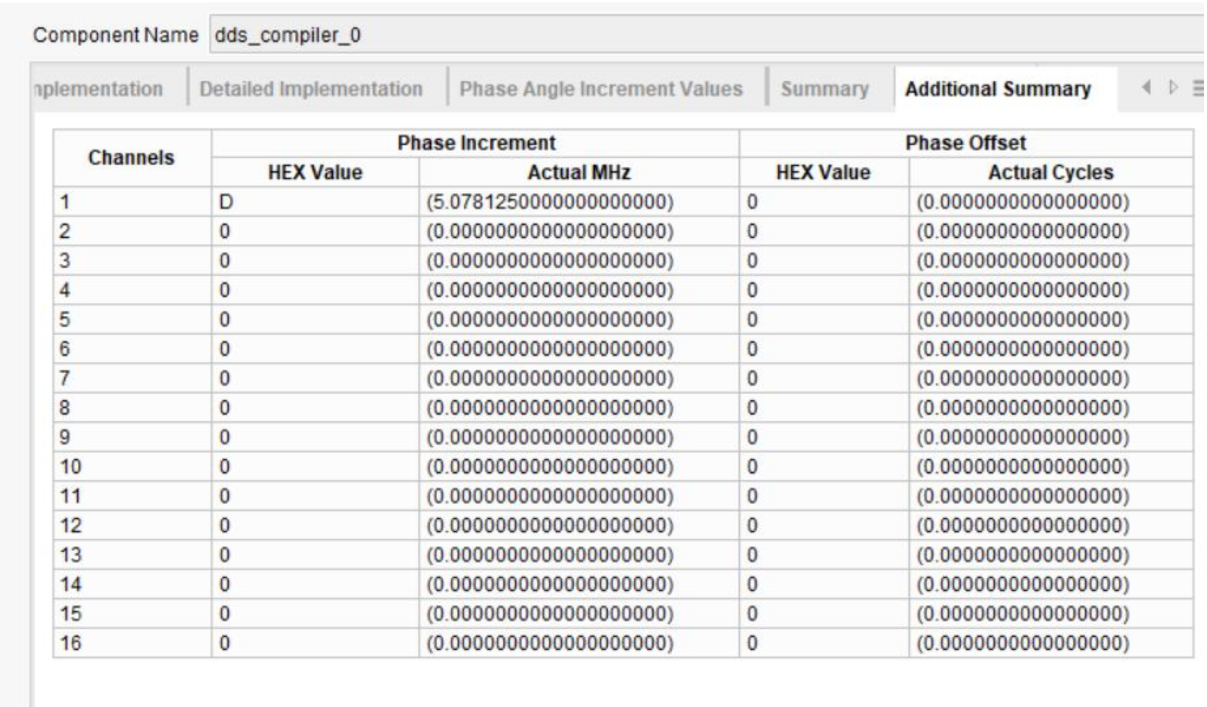

Component Name
dds_compiler_0
Configuration
Implementation
Detailed Implementation
Phase Angle Increment Values
Summary
Addit
Phase Angle Increment Values
Channel Phase Angle Increment Values (Binary)
1 1101
2 0
3 0
4 0
5 0
6 0
7 0
8 0
9 0
10 0
11 0
12 0
13 0
14 0
15 0



Component Name: dds_compiler_0

Implementation | Detailed Implementation | Phase Angle Increment Values | Summary | Additional Summary

Channels	Phase Increment		Phase Offset	
	HEX Value	Actual MHz	HEX Value	Actual Cycles
1	8	(25.000000000000000000)	0	(0.000000000000000)
2	0	(0.000000000000000000)	0	(0.000000000000000)
3	0	(0.000000000000000000)	0	(0.000000000000000)
4	0	(0.000000000000000000)	0	(0.000000000000000)
5	0	(0.000000000000000000)	0	(0.000000000000000)
6	0	(0.000000000000000000)	0	(0.000000000000000)
7	0	(0.000000000000000000)	0	(0.000000000000000)
8	0	(0.000000000000000000)	0	(0.000000000000000)
9	0	(0.000000000000000000)	0	(0.000000000000000)
10	0	(0.000000000000000000)	0	(0.000000000000000)
11	0	(0.000000000000000000)	0	(0.000000000000000)
12	0	(0.000000000000000000)	0	(0.000000000000000)
13	0	(0.000000000000000000)	0	(0.000000000000000)
14	0	(0.000000000000000000)	0	(0.000000000000000)
15	0	(0.000000000000000000)	0	(0.000000000000000)
16	0	(0.000000000000000000)	0	(0.000000000000000)



Component Name: dds_compiler_0

nplementation | Detailed Implementation | Phase Angle Increment Values | Summary | Additional Summary

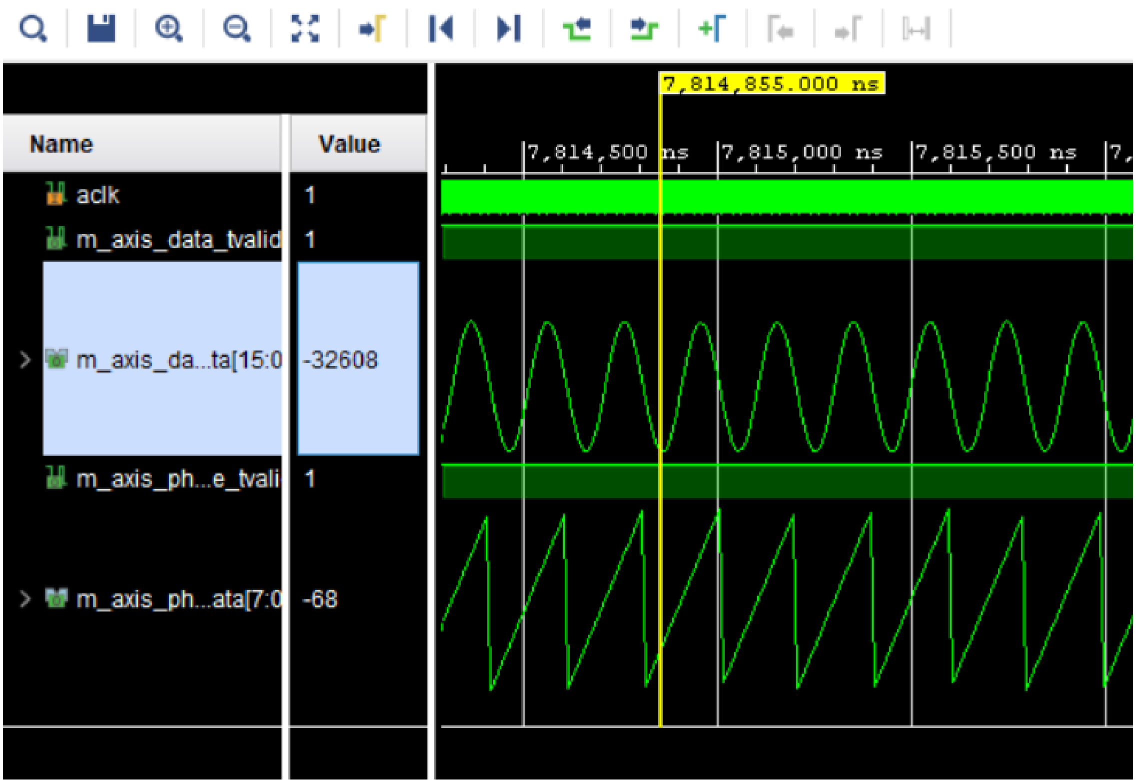
Channels	Phase Increment		Phase Offset	
	HEX Value	Actual MHz	HEX Value	Actual Cycles
1	D	(5.078125000000000000)	0	(0.000000000000000)
2	0	(0.000000000000000000)	0	(0.000000000000000)
3	0	(0.000000000000000000)	0	(0.000000000000000)
4	0	(0.000000000000000000)	0	(0.000000000000000)
5	0	(0.000000000000000000)	0	(0.000000000000000)
6	0	(0.000000000000000000)	0	(0.000000000000000)
7	0	(0.000000000000000000)	0	(0.000000000000000)
8	0	(0.000000000000000000)	0	(0.000000000000000)
9	0	(0.000000000000000000)	0	(0.000000000000000)
10	0	(0.000000000000000000)	0	(0.000000000000000)
11	0	(0.000000000000000000)	0	(0.000000000000000)
12	0	(0.000000000000000000)	0	(0.000000000000000)
13	0	(0.000000000000000000)	0	(0.000000000000000)
14	0	(0.000000000000000000)	0	(0.000000000000000)
15	0	(0.000000000000000000)	0	(0.000000000000000)
16	0	(0.000000000000000000)	0	(0.000000000000000)

Testbench

```
`timescale 1ns / 1ps

module tb_Sine_wave;

reg clk;

reg rst;
```

```
wire clk_out;
dds_compiler_0 uut (
   .aclk(clk),
   .m_axis_data_tvalid(),
   .m_axis_data_tdata(),
   .m_axis_phase_tvalid(),
   .m_axis_phase_tdata()
);
always #5 clk = ~clk;
initial begin
   clk = 0;
   rst = 1;
   #20 rst = 0;
   #100;
   $stop;
end
endmodule
```

Output



Result

A DDS system was successfully implemented using the DDS Compiler IP core for the **generation of a 5 MHz signal**. The simulated sine outputs were continuous and periodic. The obtained output frequency was approximately **5.08 MHz**, which closely matches the desired 5 MHz signal, confirming the correct selection of the phase increment value and proper functioning of the DDS.

(b) Generation of a cosine wave with a frequency of 10MHz

Aim
To design and simulate a Direct Digital Synthesizer (DDS) using the DDS Compiler IP core for the generation of a 10 MHz cosine wave signal.

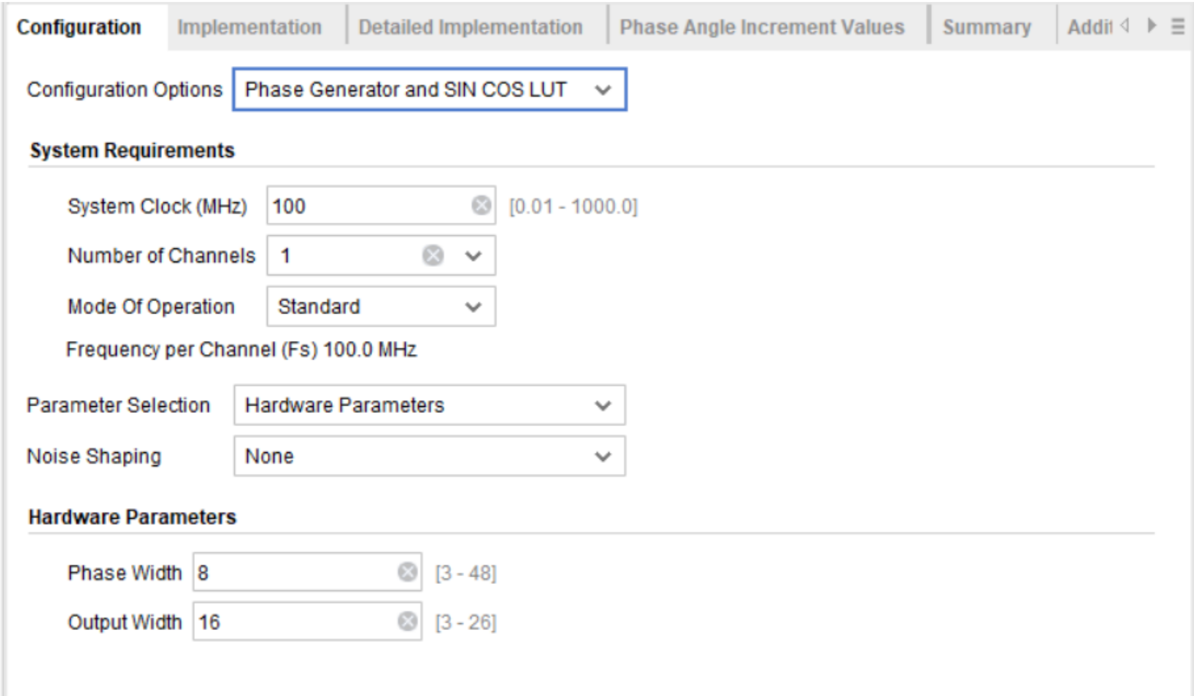
Software Required
Xilinx Vivado Design Suite
Vivado Simulator
Vivado IP Catalog

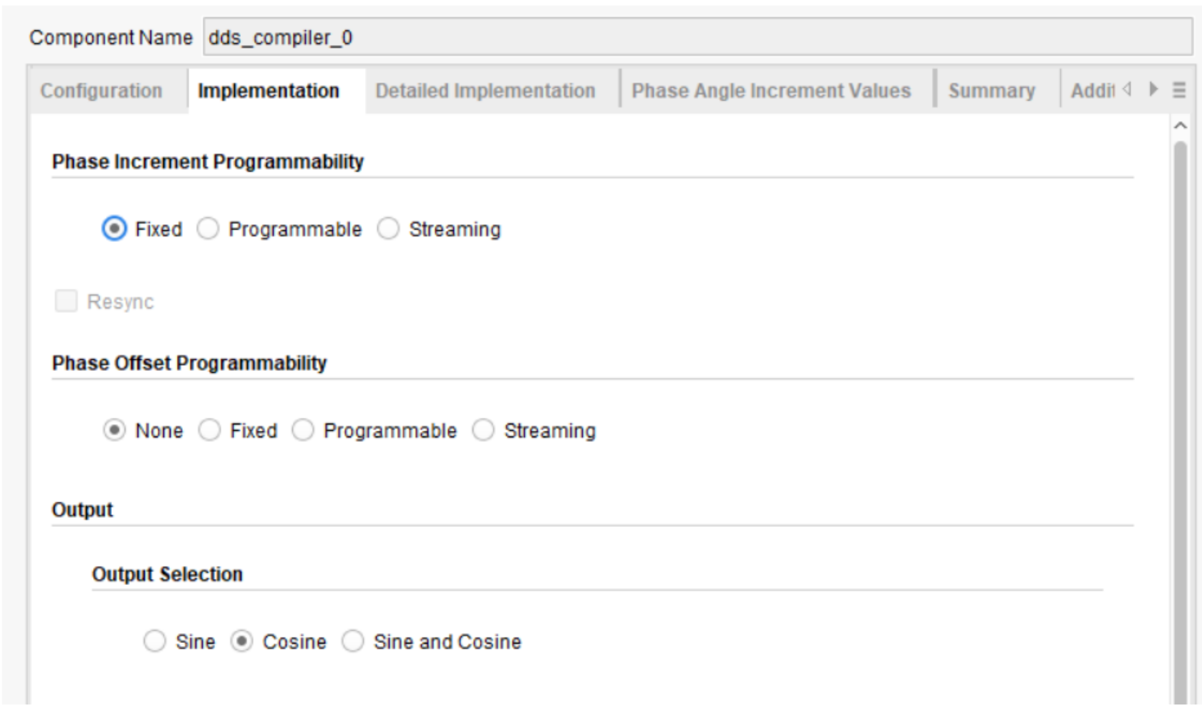
Procedure

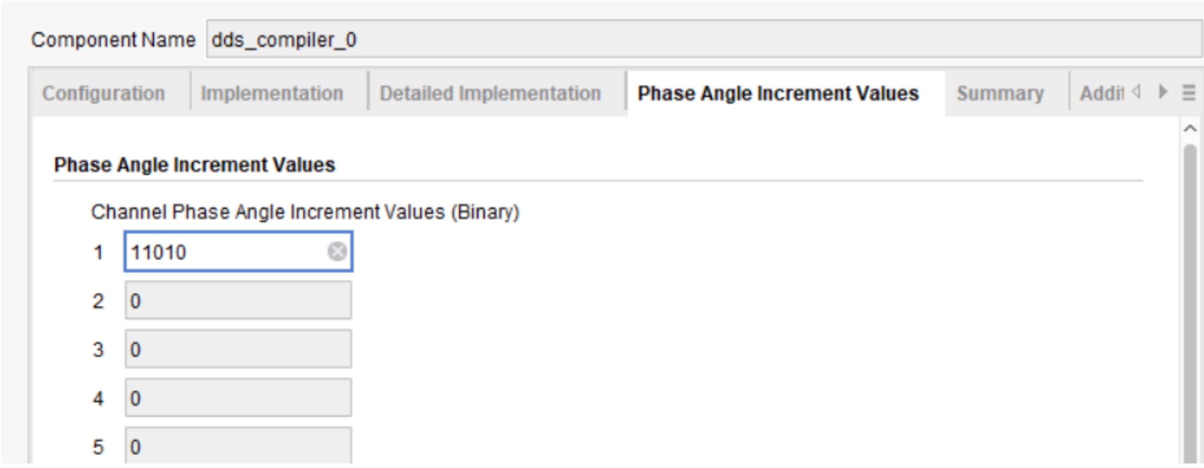
Open Xilinx Vivado and create a new RTL project.
Select the target FPGA device/board.
From the IP Catalog, add DDS Compiler (dds_compiler_0).
Configure the IP as follows:
Configuration: Phase Generator and SIN COS LUT
System clock: 100 MHz
Channels: 1
Mode: Standard
Phase width: 8 bits
Output width: 16 bits
Noise shaping: None
Go to the Phase Angle Increment Values tab and set:
Phase Increment = 26
Generate the IP output products.
Write a testbench to apply clock and reset and to capture the cosine output.
Run behavioral simulation and observe the cosine waveform.

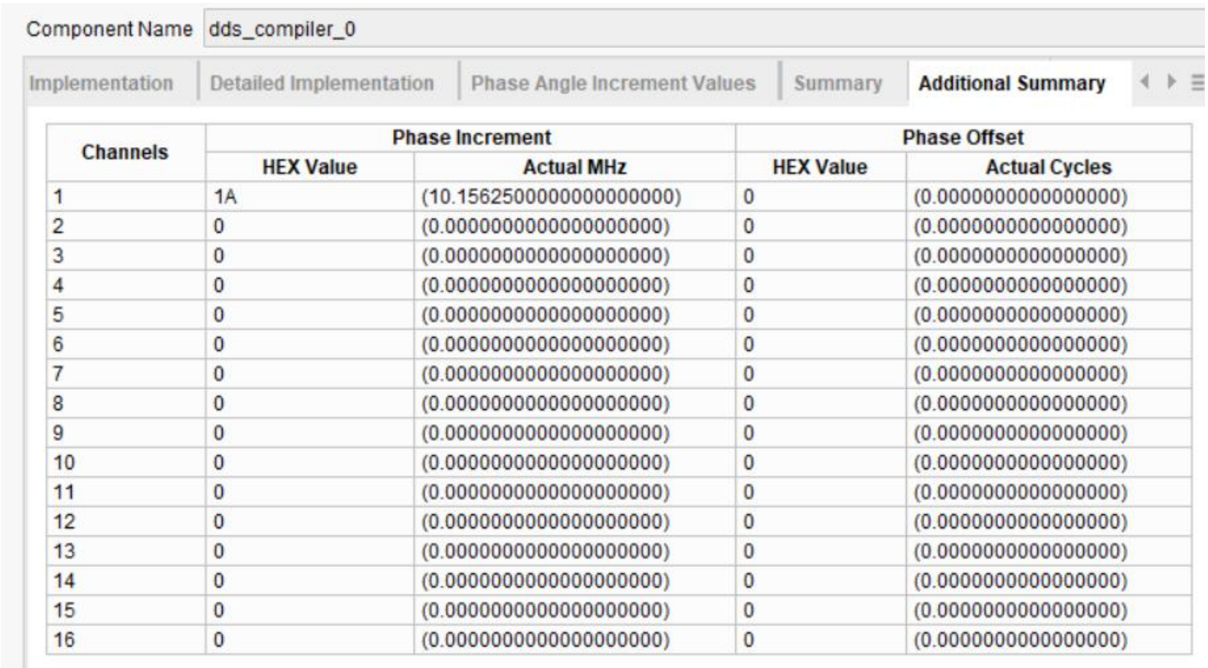
Frequency calculation
f_out = (Phase Increment × f_clk) / 2^(Phase width)

f_out = (26 × 100 MHz) / 256 ≈ 10.16 MHz


Configuration
Implementation
Detailed Implementation
Phase Angle Increment Values
Summary
Addit
Configuration Options
Phase Generator and SIN COS LUT
System Requirements
System Clock (MHz)
100
[0.01 - 1000.0]
Number of Channels
1
Mode Of Operation
Standard
Frequency per Channel (Fs) 100.0 MHz
Parameter Selection
Hardware Parameters
Noise Shaping
None
Hardware Parameters
Phase Width
8
[3 - 48]
Output Width
16
[3 - 26]


Component Name
dds_compiler_0
Configuration
Implementation
Detailed Implementation
Phase Angle Increment Values
Summary
Addit
Phase Increment Programmability
Fixed
Programmable
Streaming
Resync
Phase Offset Programmability
None
Fixed
Programmable
Streaming
Output
Output Selection
Sine
Cosine
Sine and Cosine



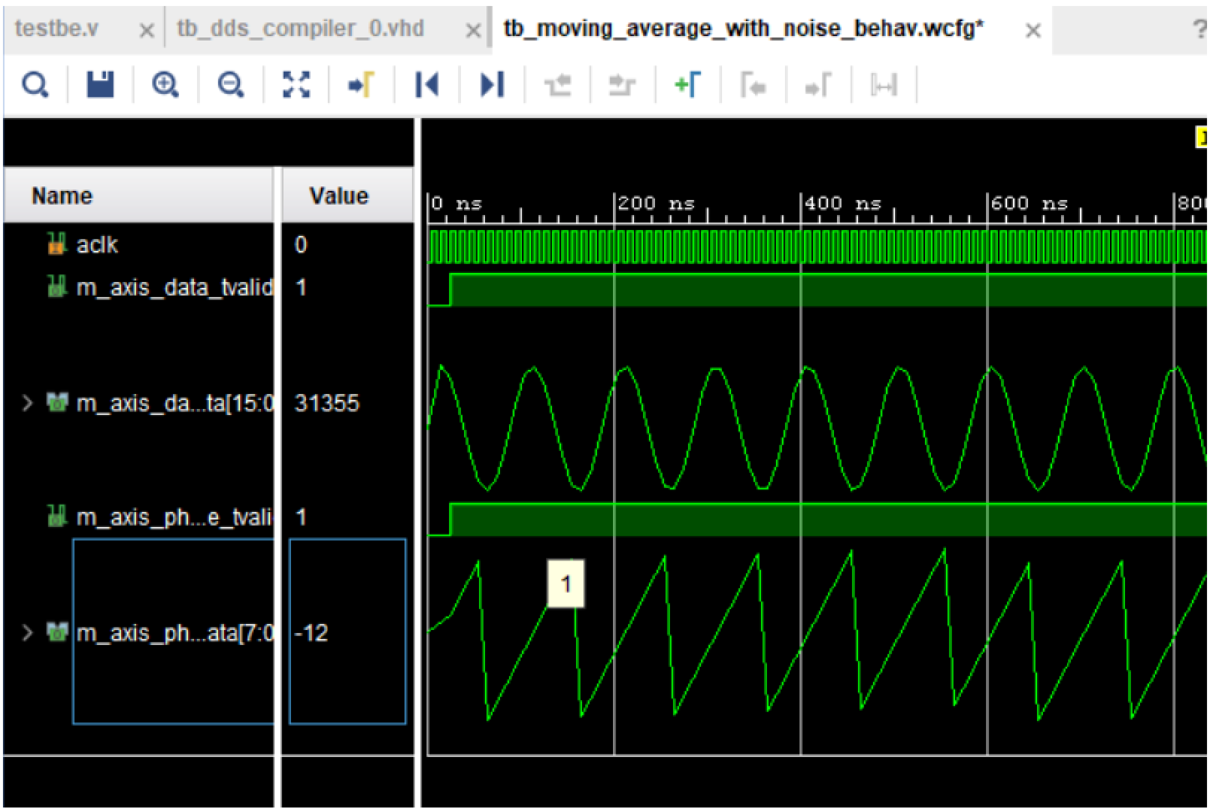


Channels	Phase Increment		Phase Offset	
	HEX Value	Actual MHz	HEX Value	Actual Cycles
1	1A	(10.156250000000000000)	0	(0.0000000000000000)
2	0	(0.000000000000000000)	0	(0.0000000000000000)
3	0	(0.000000000000000000)	0	(0.0000000000000000)
4	0	(0.000000000000000000)	0	(0.0000000000000000)
5	0	(0.000000000000000000)	0	(0.0000000000000000)
6	0	(0.000000000000000000)	0	(0.0000000000000000)
7	0	(0.000000000000000000)	0	(0.0000000000000000)
8	0	(0.000000000000000000)	0	(0.0000000000000000)
9	0	(0.000000000000000000)	0	(0.0000000000000000)
10	0	(0.000000000000000000)	0	(0.0000000000000000)
11	0	(0.000000000000000000)	0	(0.0000000000000000)
12	0	(0.000000000000000000)	0	(0.0000000000000000)
13	0	(0.000000000000000000)	0	(0.0000000000000000)
14	0	(0.000000000000000000)	0	(0.0000000000000000)
15	0	(0.000000000000000000)	0	(0.0000000000000000)
16	0	(0.000000000000000000)	0	(0.0000000000000000)

Testbench program

```
`timescale 1ns / 1ps

module tb_Cosine_wave;

reg clk;

reg rst;

wire clk_out;

dds_compiler_0 uut (

   .aclk(clk),

   .m_axis_data_tvalid(),
```

```
    .m_axis_data_tdata(),
    .m_axis_phase_tvalid(),
    .m_axis_phase_tdata()
);
always #5 clk = ~clk;
initial begin
    clk = 0;
    rst = 1;
    #20 rst = 0;
    #100;
    $stop;
end
endmodule
```



Result
The DDS Compiler IP core was successfully configured to generate a cosine waveform. The simulated output frequency was approximately 10.16 MHz, which closely matches the desired 10 MHz signal, confirming correct DDS operation and phase increment selection.